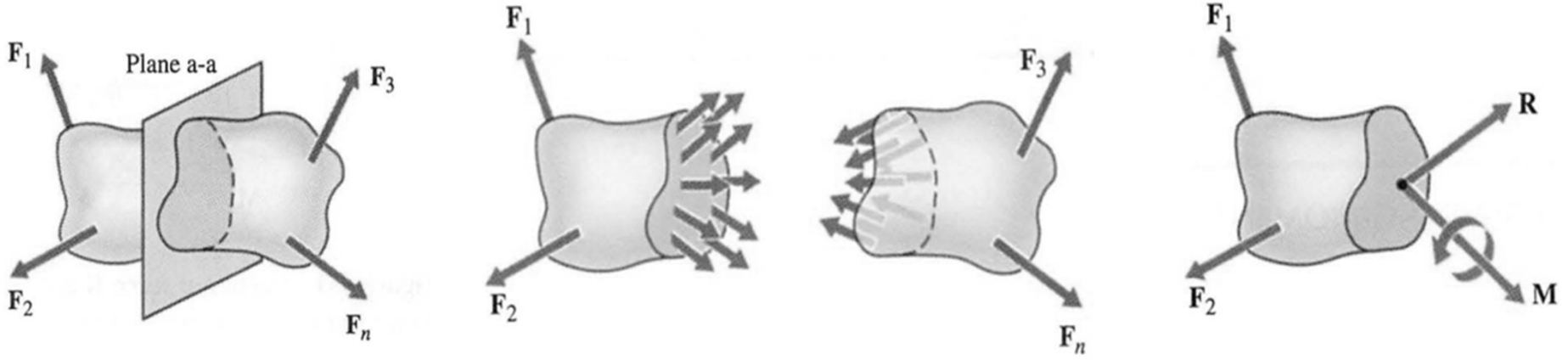
LECTURE 2

Internal Reactions

Internal Reactions

- Internal reactions (forces and moments) are generated to balance and react to the external loads on a body.



- To determine the internal forces on the cut section of a structural member;
  - determine all external and reaction forces or torques that will ensure static equilibrium of the structural member.
  - perform an equilibrium analysis on the cut section of interest of the structural member.

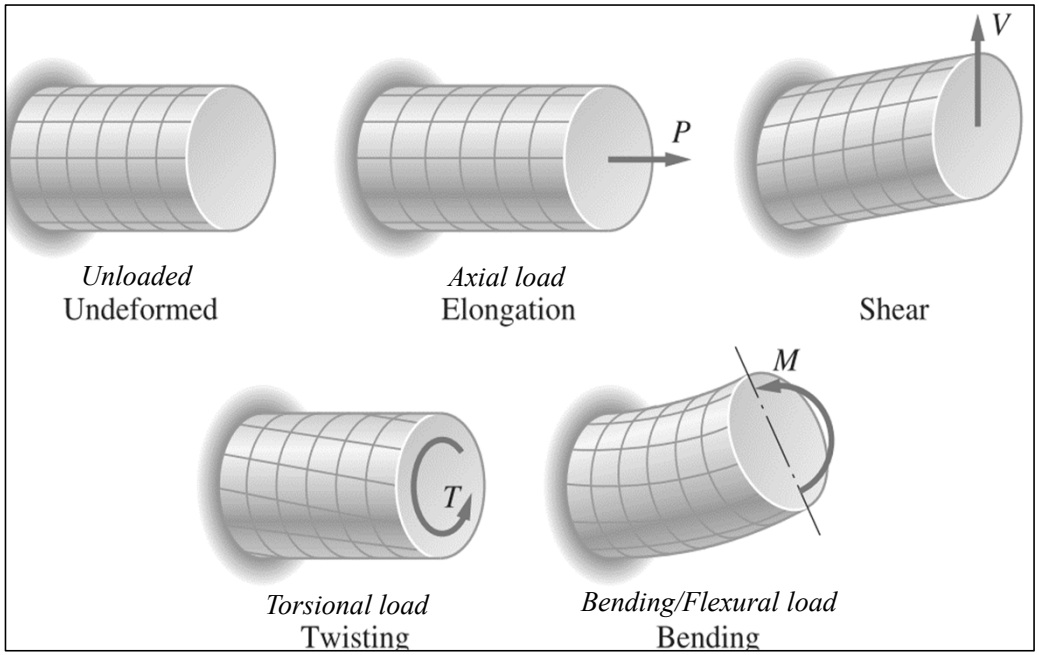
Stress and Strain

- Stress is a quantity that accounts for the intensity and distribution of internal reactions (forces and moments).

- Strain is the quantity measures the intensity of the deformation in a stressed body.

- External loads are also of several types;
  - Axial loads
  - Torsional loads
  - Bending loads

- Stresses depend on the type of external load;
  - Tensile stress and compressive stress
  - Bending/Flexural stress
  - Shear Stress
  - Thermal stress



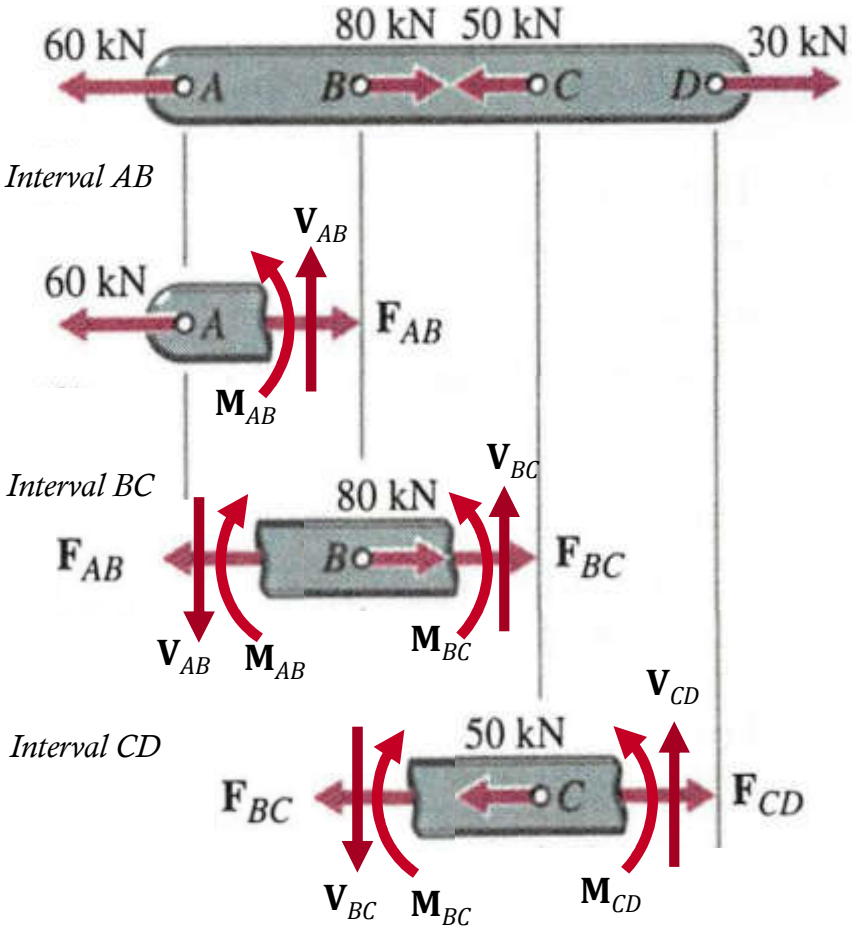
Source: Mechanics of Materials, Pytel and Kiusalaas

Internal Reactions

Example

Determine the axial forces transmitted by cross sections in intervals AB, BC, and CD of the bar shown below. Also determine the stress at each interval if the bar has a constant cross section area of 100 cm^2.





$\overset{+}{\rightarrow}\sum F = 0: \; -60\text{ kN} + F_{AB} = 0$

$\Rightarrow F_{AB} = 60\text{ kN}$

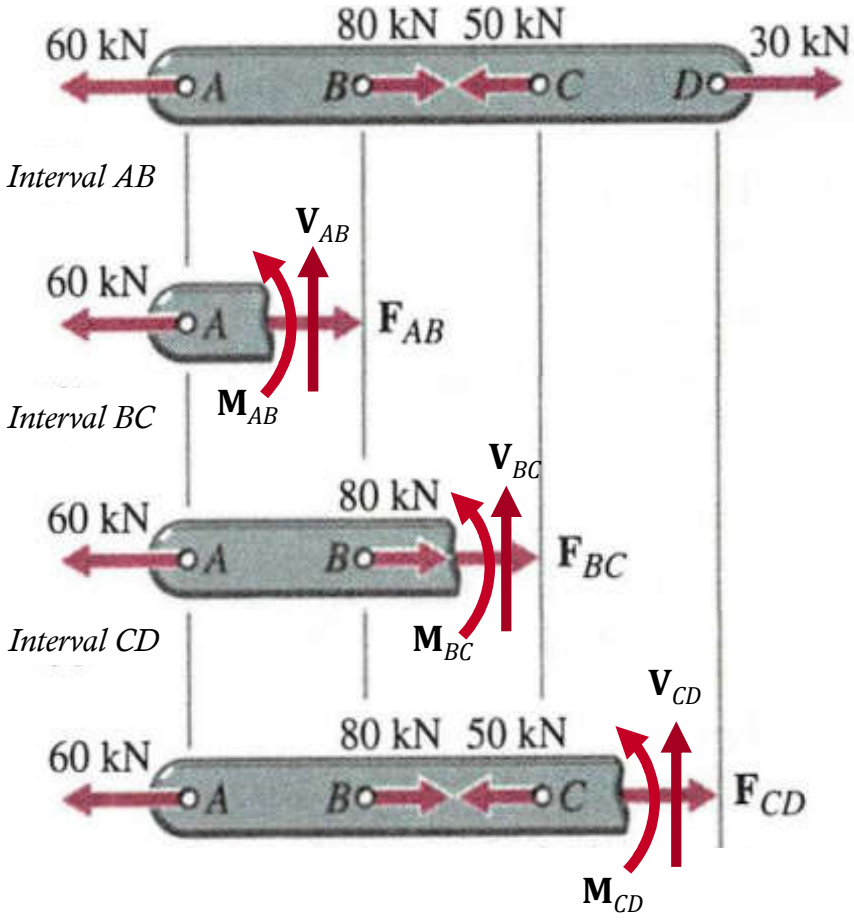
$\overset{+}{\rightarrow}\sum F = 0: \; -60\text{ kN} + 80\text{ kN} + F_{BC} = 0$

$\Rightarrow F_{BC} = -20\text{ kN}$

$F_{CD} = 30\text{ kN}$

Internal Reactions

Alternatively;



$$\overset{+}{\rightarrow}\sum F = 0: \; -60\text{ kN} + F_{AB} = 0$$
$$\Rightarrow F_{AB} = 60\text{ kN}$$

$$\overset{+}{\rightarrow}\sum F = 0: \; -60\text{ kN} + 80\text{ kN} + F_{BC} = 0$$
$$\Rightarrow F_{BC} = -20\text{ kN}$$

$$F_{CD} = 30\text{ kN}$$

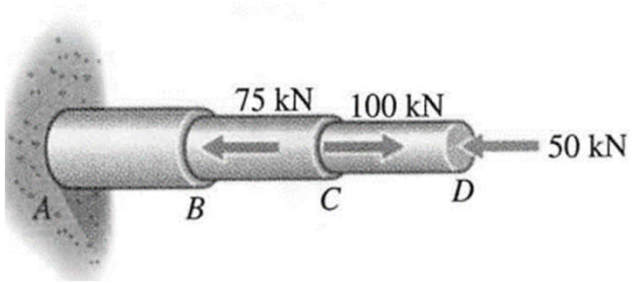
Internal Reactions

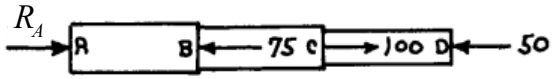
Example:

A steel shaft with a fixed joint to a wall has axial loads applied as shown below.

a. Determine the axial forces transmitted by transverse cross-sections in intervals AB, BC, and CD of the bar.

b. Sketch the internal force diagram for the shaft.



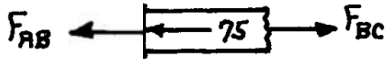


$$\overset{+}{\rightarrow} \sum F = 0 : R_A - 75\ \text{kN} + 100\ \text{kN} - 50\ \text{kN} = 0$$

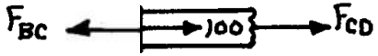
$$\Rightarrow R_A = 25\ \text{kN}$$



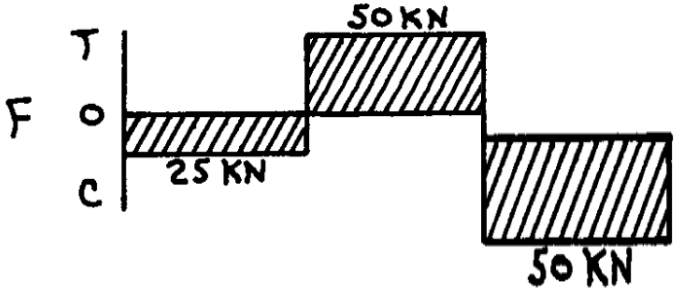
$$F_{AB} = -25\ \text{kN}$$



$$F_{BC} = 50\ \text{kN}$$

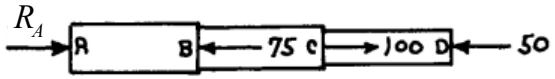


$$F_{CD} = -50\ \text{kN}$$



Internal Reactions

Alternatively;



$$\overset{+}{\rightarrow} \sum F = 0 : R_A - 75\ \text{kN} + 100\ \text{kN} - 50\ \text{kN} = 0$$

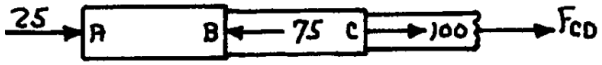
$$\Rightarrow R_A = 25\ \text{kN}$$



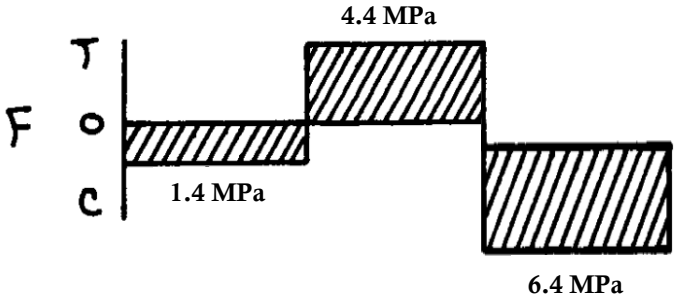
$$F_{AB} = -25\ \text{kN}$$



$$F_{BC} = 50\ \text{kN}$$



$$F_{CD} = -50\ \text{kN}$$



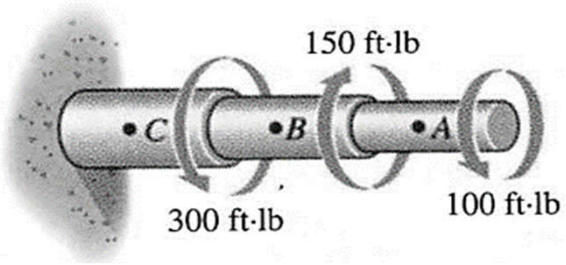
Internal Reactions

Example

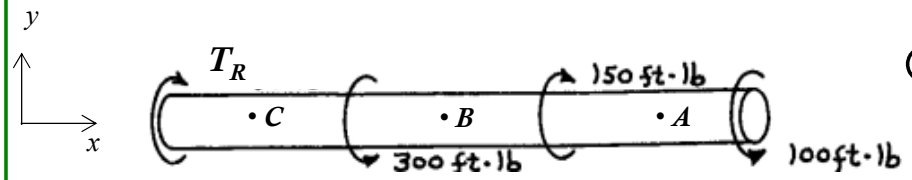
For the steel shaft shown;

a) Determine the torque transmitted by the transverse cross sections at points A, B, and C.

b) Sketch a torque diagram for the shaft.

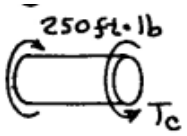


For support reactions on the shaft, recall

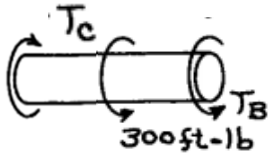


$\circlearrowleft \sum M_x = 0 : T_R - 300 + 150 - 100 = 0$

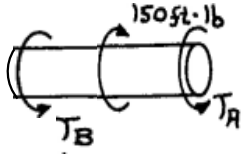
$\therefore \; T_R = 250 \text{ lb.ft} \; \circlearrowright$



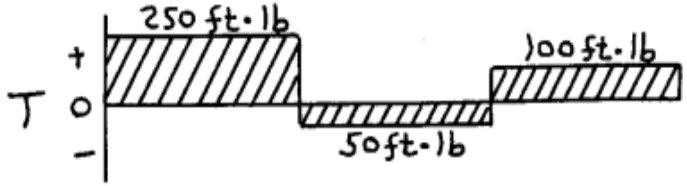
$\circlearrowleft \sum M = 0 : T_C = 250 \text{ ft.lb} \; \circlearrowright$



$\circlearrowleft \sum M = 0 : T_B = 50 \text{ ft.lb} \; \circlearrowright$



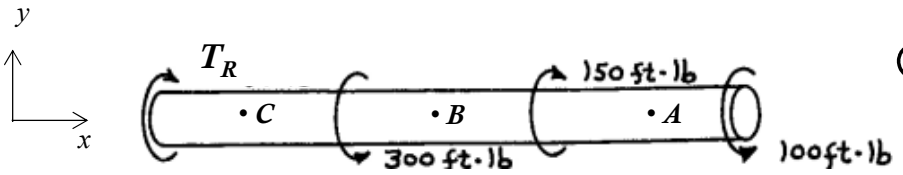
$\circlearrowleft \sum M = 0 : T_A = 100 \text{ ft.lb} \; \circlearrowright$



Torsional Loading: Internal Forces

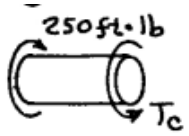
Alternatively;

For support reactions on the shaft, recall

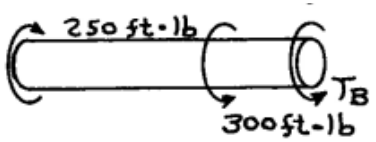


$\circlearrowleft \sum M_x = 0 \; : T_R - 300 + 150 - 100 = 0$

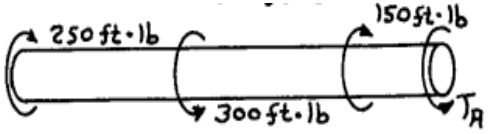
$\therefore \; T_R = 250 \text{ lb.ft} \; \circlearrowright$



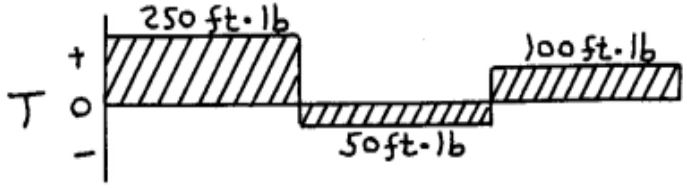
$\circlearrowleft \sum M = 0 \; : T_C = 250 \text{ ft.lb} \; \circlearrowright$



$\circlearrowleft \sum M = 0 \; : T_B = 50 \text{ ft.lb} \; \circlearrowright$



$\circlearrowleft \sum M = 0 \; : T_A = 100 \text{ ft.lb} \; \circlearrowright$



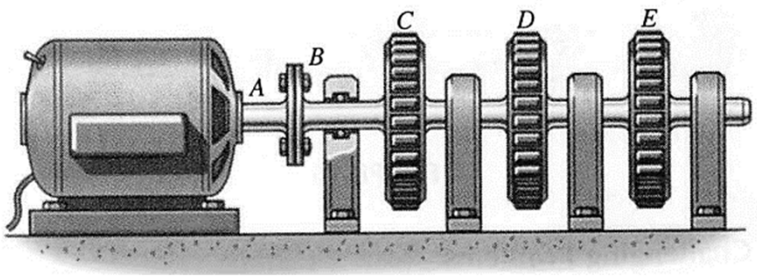
Internal Reactions

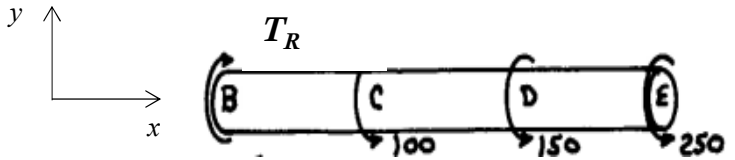
Example

The motor shown below supplies a torque of 500 Nm to shaft BCDE. The torques removed at gears C, D, and E are 100 Nm, 150 Nm, and 250 Nm respectively.

a) Determine the torques transmitted by the transverse cross sections in intervals BC, CD, and DE and sketch a torque diagram

b) In what section does the maximum shear stress occur?

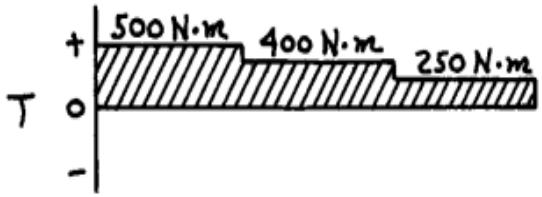




$\circlearrowleft \sum M_x = 0 \; : T_R - 100 - 150 - 250 = 0$

$\therefore \; T_R = 500 \text{ N.m} \circlearrowright$

$\circlearrowleft \sum M = 0 \; : T_{BC} = 500 \text{ N.m} \circlearrowright$



$\circlearrowleft \sum M = 0 \; : T_{CD} = 400 \text{ N.m} \circlearrowright$

$\circlearrowleft \sum M = 0 \; : T_{DE} = 250 \text{ N.m} \circlearrowright$

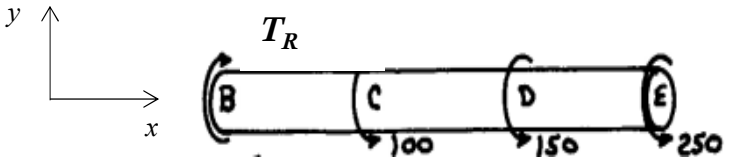
Internal Reactions

Example

The motor shown below supplies a torque of 500 Nm to shaft BCDE. The torques removed at gears C, D, and E are 100 Nm, 150 Nm, and 250 Nm respectively.

a) Determine the torques transmitted by the transverse cross sections in intervals BC, CD, and DE and sketch a torque diagram

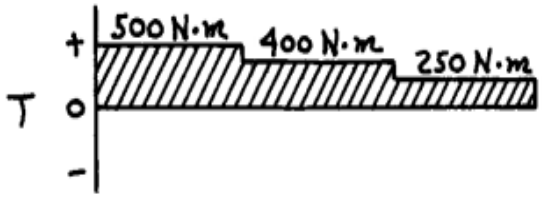
b) In what section does the maximum shear stress occur?



$$\circlearrowleft \sum M_x = 0 \; : T_R - 100 - 150 - 250 = 0$$

$$\therefore \; T_R = 500 \text{ N.m} \circlearrowright$$

$$\circlearrowleft \sum M = 0 \; : T_{BC} = 500 \text{ N.m} \circlearrowright$$



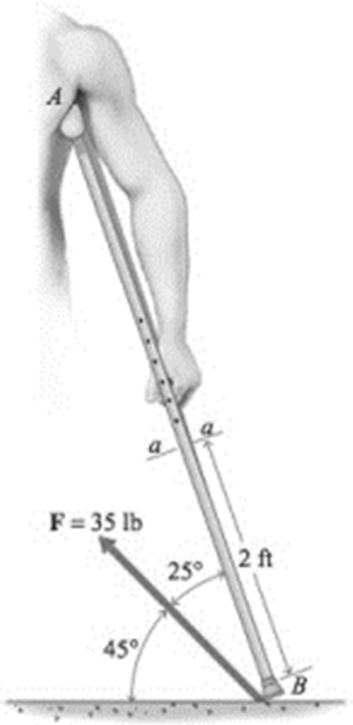
$$\circlearrowleft \sum M = 0 \; : T_{CD} = 400 \text{ N.m} \circlearrowright$$

$$\circlearrowleft \sum M = 0 \; : T_{DE} = 250 \text{ N.m} \circlearrowright$$

Internal Reactions

Example:

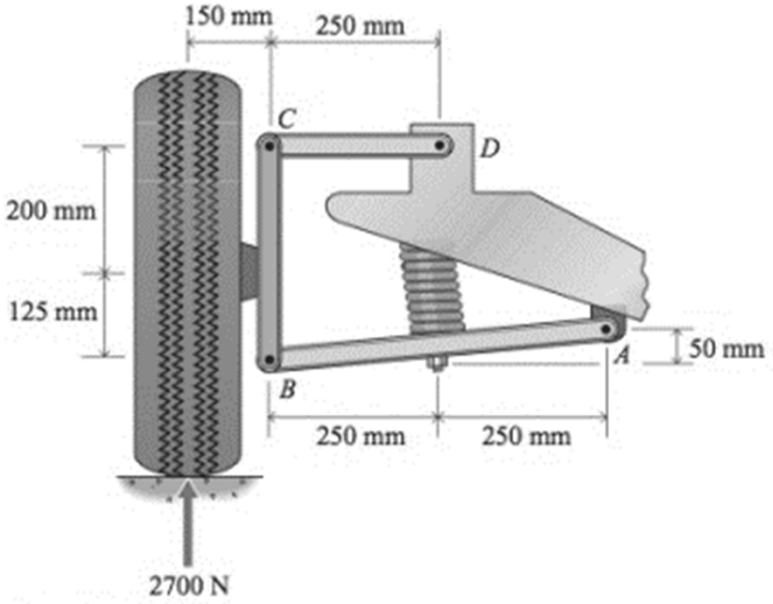
The reaction between a crutch and the ground is 35 lb, as shown. Determine the internal forces acting at section a-a.



Normal force = 31.7 lb (C)
Shear Force = 14.79
Moment (about the cut) = 29.6 lb.ft (CW)

Example:

The front-wheel suspension of an automobile is shown below. The pavement exerts a vertical force of 2700 N on the tire. Determine the internal forces on a section perpendicular to and midway between C and D. (Hint: Member CD is a 2-force member)



Normal force = 1246 N(C)
Shear Force = 0
Moment = 0